

OT Disaster Recovery Objectives

The overall objective for creating an OT DR plan is to ensure that people, processes, and underlying technology are ready to respond effectively to right-size an ICS environment in the event of a disaster. Specifically, a DRP should cover how to:

- Manage the crisis in an organized and effective manner, reducing recovery time
- Limit the magnitude or impact of a cyber attack to the various production locations
- Create a recovery plan that, under various impact scenarios, reestablishes the plant's mission within the optimal recovery time

Plans Included in Business Continuity Management

An OT DRP is a subset of the overall Business Continuity Management (BCM) set of plans and capabilities.

- **Business Continuity Plan (BCP):** Provides procedures for sustaining essential business operations while recovering from a significant disruption.
- **Continuity Of Operations Plan (COOP):** Provides procedures and capabilities to sustain a plant’s essential functions at an alternate site or when constrained due to failed (sub)systems for a predefined recovery period.
- **Crisis Communications Plan (CCP):** Provides procedures for disseminating status reports to personnel and the public.
- **Occupant Emergency Plan (OEP):** Provides coordinated procedures for minimizing loss of life or injury and protecting property from damage in response to a physical threat.

Plans Included in Business Continuity Management, cont.

- **Incident Response Plan (IRP):** This plan provides strategies to detect, respond to, and limit the consequences of malicious cyber incidents.
- **Information System Contingency Plan (ICSP):** Provides procedures and capabilities for recovering an information system.
- **The Disaster Recovery Plan (DRP):** Provides detailed procedures to facilitate the recovery of capabilities at the local or alternate site.

OT Recovery Phases

Activation and Notification: Initial actions to take once a system disruption or outage has been detected, which is typically listed out in the OT IRP.

- Describe the notification of recovery personnel
- Define the outage assessment process
- Define the activation process and team mobilization

Recovery: Actions to take after a system has been made safe and cyber-ready for recovery.

- Define recovery strategies
- Define recovery sequence and activities/procedures per strategy
- Define escalation procedures
- Define backup repository, backup media selection, and transport

Reconstitution: Final data-driven actions to ensure recovered systems have the most recent operational parameters before a cyber-physical system restart.

- Deconflict concurrent data processing in certain scenarios
- Validate recovered data to ensure the latest configurations and datasets are loaded
- Validate the correct operation and functionality of the system after recovery



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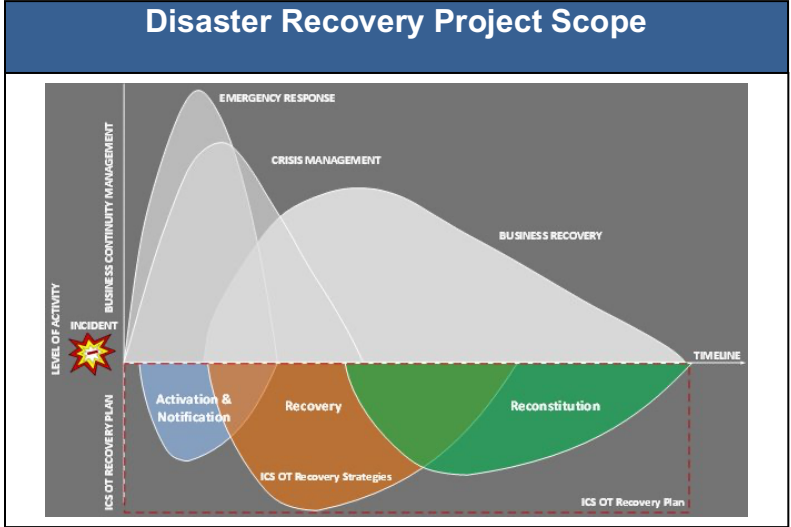
OT Disaster Recovery Quick Start Guide v1.0

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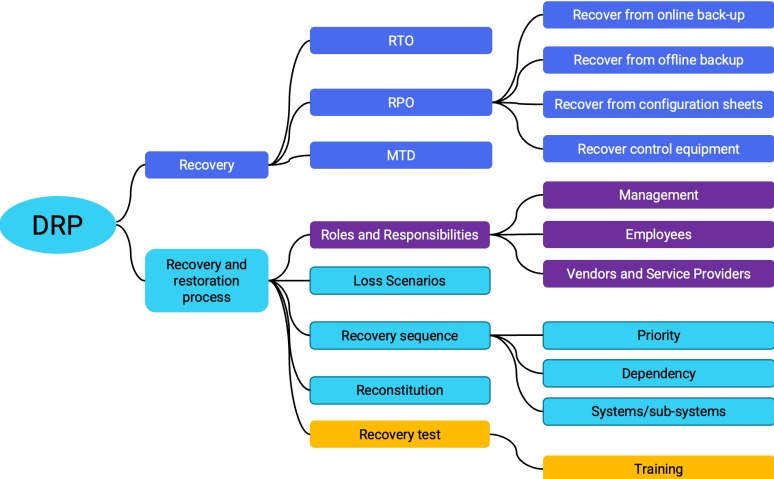
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This tri-fold cheat sheet provides an overview of the primary components of creating a Disaster Recovery plan in Industrial Control System (ICS) and Operational Technology environments (OT)

Use this guide to assess the recovery phase of your Incident Response program (IR) and refresh or develop an ICS/OT DRP within your organization.



Major Elements of a DRP



- **Recovery Time Objective (RTO):** RTO defines the maximum amount of time that a system resource/component can remain unavailable before there is an unacceptable impact on other system resources/components, supported automation processes, and the MTD.
- **Recovery Point Objective (RPO):** The RPO represents the point in time, prior to a disruption or system outage, to which mission/business data can be recovered after an outage. Unlike RTO, RPO is not considered part of the MTD.
- **Maximum Tolerable Downtime (MTD):** The MTD is the total amount of time the business is willing to accept for a production (sub)process outage or disruption.

The MTD is the key parameter for recovery. It drives the recovery strategy, the level of detail for the recovery procedures, and their scope and content.

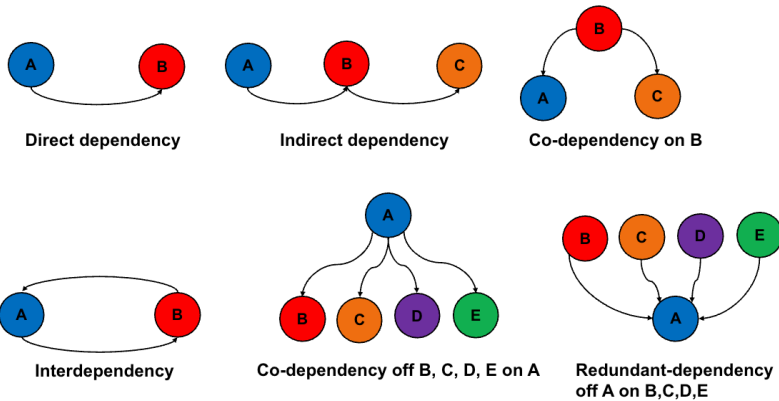
Recovery Strategy and Process

Recovery Strategy:
Selection of loss scenarios that will invoke the crisis and emergency management process. Information about the ICS systems' optimal recovery sequence for each of the identified scenarios to optimize reconstitution time.

Recovery Process:
Administrative procedures are essential to prevent disorganized and confused responses to Incidents and to limit an incident's potential for damaging industrial control systems.

Recovery Elements: Recovery Scenarios, Teams, Mandates and Responsibilities, Invocation, Mobilization, and Communication, Recovery Capabilities, Materials and Prioritization, Disaster Recovery Testing.

Dependency Analysis



- Direct dependency on A
- Indirect dependency on A
- Co-dependency on B
- Interdependency
- Co-dependency on A
- Redundant dependency off A

Disaster Scenarios

What constitutes the disaster recovery criteria? What kind of incident meets disaster criteria?

- **SD – 3:** Anomaly, local incident, impaired support function, no process downtime.
- **SD – 2:** Noteworthy cyber incidents, local incident, impaired essential function, loss of support function, lost interconnectedness between locations, process downtime less than RTO.
- **SD – 1:** Extensive incidents, single site/multiple functions impacted, manual operation possible. Loss of connectivity, loss of performance, loss of ability to perform, Loss of essential function. Downtime less than MTD.
- **SD-0:** Major or widespread incident, multi-site impact, multiple essential functions impacted, manual operation not possible. Loss of connectivity, loss of data exchange between essential functions. Downtime $\geq$ MTD.

Disasters are considered loss scenarios that result in a shutdown lasting more than MTD of recovery time. Shutdown 0 (SD-0) scenarios are considered disasters that invoke the disaster recovery process or Emergency Response Team.

Recovery Summary

It's important to remember that large disaster recovery efforts are much more than "restoration from backups."

In the overall recovery scope, remember:

- Automation Functions
- Recovery Priorities
- Dependencies
- Data reconstitution of the automation systems